Model selection: three search strategies

Pick a good subset of the predictors $X_1, \ldots, X_p$.

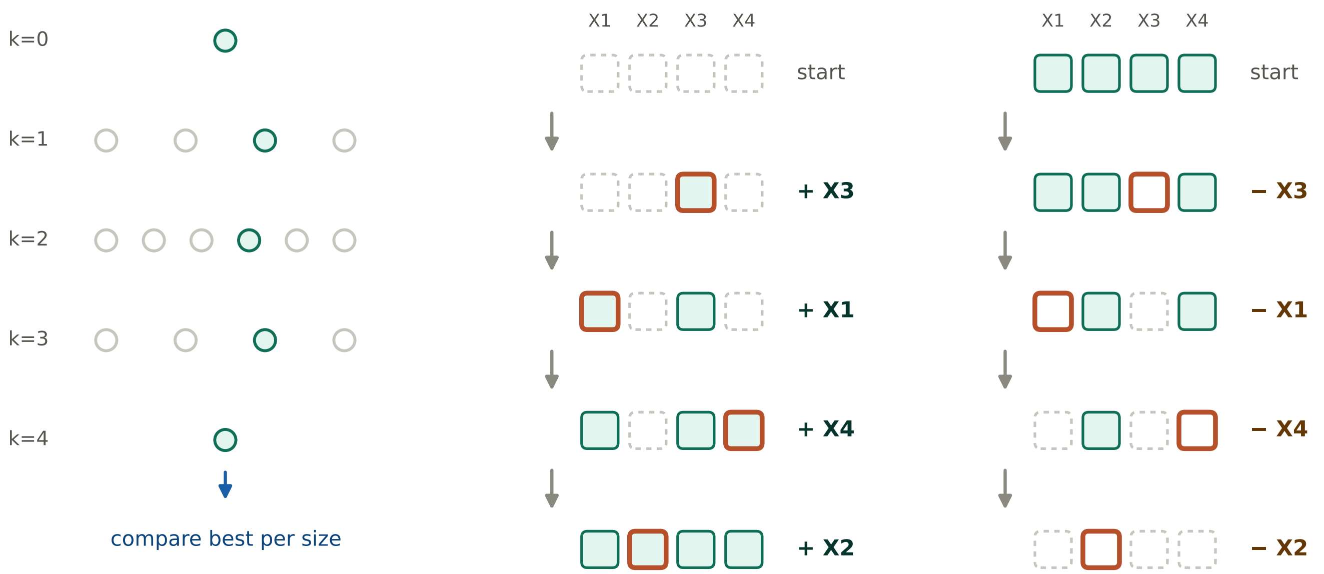
Setup

A model uses a subset $S \subseteq \{X_1, \ldots, X_p\}$. Score each model with a criterion C.

Use AIC, BIC, or cross-validation error (a lower C is better). All three methods share this score.

	Best subset Check every subset.	**Forward selection** Grow from nothing.	**Backward selection** Prune from everything.
Start	every subset	$S = \{\ \}$ (empty)	$S = \{X_1, \ldots, X_p\}$
The move	Fit all subsets. Keep the best at each size k.	Add the predictor that improves C the most. Repeat.	Drop the predictor that hurts C the least. Repeat.

Search path



fits 2^p models in total

	Best subset	Forward selection	Backward selection
Models fit	2^p	$1 + \frac{p(p+1)}{2}$	$1 + \frac{p(p+1)}{2}$ (needs $n > p$)

Then choose Pick the model with the best C among the candidates each method produced.

Why the strategies differ

Best subset checks every model, so it finds the best subset for each size. But 2^p grows fast.
Forward and backward are greedy. They are cheap, but they follow one path, so they can miss the best subset.
Backward needs more observations than predictors ($n > p$) to fit the full model at the start.

- predictor in the model
- predictor left out
- changed this step
- best model of that size